

# MANAV SHARMA

MERN Stack Developer · Backend & System Design

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## EDUCATION

### National Institute of Technology (NIT) Bhopal

2023 – 2027

B.Tech, Electronics & Communication Engineering · **CGPA: 7.06/10**

Coursework: DBMS, Operating Systems · Class XII (CBSE): 77.0% · Class X (CBSE): 92.0%

## TECHNICAL SKILLS

|                  |                                                                                                |
|------------------|------------------------------------------------------------------------------------------------|
| Languages:       | C++, JavaScript (ES6+), Python                                                                 |
| Backend:         | Node.js, Express.js, REST APIs, Socket.io, JWT, OAuth 2.0, Passport.js, RBAC, Bcrypt, Razorpay |
| Data & Queues:   | MongoDB, Mongoose, PostgreSQL, MySQL, Redis, BullMQ                                            |
| Infra & DevOps:  | Cloudflare R2, Cloudinary, PM2, GitHub Actions, CI/CD, Vercel, Render, FFmpeg, HLS             |
| Frontend:        | React.js, Redux Toolkit, Tailwind CSS, Three.js, GSAP, Framer Motion, Lenis                    |
| Testing & Tools: | K6, Artillery, Autocannon, Clinic.js, Lighthouse, Git, Postman, npm                            |

## PROJECTS

### NexForge — Self-Hosted PaaS Platform

[Live](#) · [GitHub](#) · [npm](#)

[Node.js](#), [BullMQ](#), [Redis](#), [Socket.io](#), [React](#), [GitHub OAuth](#), [PM2](#), [Razorpay](#)

- Architected a self-hosted PaaS with build, deployment, and telemetry APIs — JWT auth, GitHub OAuth, artifact compression, and zero-downtime rollback; shipped an npm CLI with **1,076+ weekly downloads**.
- Built a concurrent CI/CD pipeline on BullMQ + Redis with PM2 and symlink-based artifact swapping, PAT-authenticated CLI uploads, and Socket.io real-time build logs.
- Monetized with Razorpay subscriptions — plan-gated limits enforced across dashboard and CLI, webhook-verified payments, and atomic free-tier quota checks.
- Load-tested with Autocannon, Artillery, and K6: **1M+ requests at 2,802 req/s (100% success, P99: 193ms)**, zero errors; Clinic.js across **42K+ samples** confirmed zero leaks and a stable event loop; **Lighthouse 99** (FCP: 0.5s, LCP: 0.8s, CLS: 0).

### Adaptive Bitrate HLS Video Streaming Platform

[Live](#) · [GitHub](#)

[React](#), [Node.js](#), [FFmpeg](#), [BullMQ](#), [Redis](#), [Cloudflare R2](#), [MongoDB](#), [Razorpay](#)

- Built an adaptive-bitrate HLS engine (1080p/720p/480p, multi-audio, subtitles) covering upload, ffprobe analysis, R2 chunk storage, and delivery, with live transcode progress over Socket.io.
- FFmpeg split-filter pipeline encodes all renditions in one pass — **50%+ CPU saved** and **~45% faster** (4h to 2.25h) on 512MB servers, with auto-generated MP4 fallbacks and thumbnail sprites.
- Offloaded CPU-heavy transcoding to BullMQ workers with HLS chunks on Cloudflare R2, keeping the API path light.
- Razorpay tiers gate max resolution and downloads; load-tested with Autocannon, Artillery, and K6 across **1,000 VUs** — **187K+ requests at 480 req/s (100% success)**, zero dropped requests; **Lighthouse 90** (FCP: 0.6s, CLS: 0).

### Crochella — Multi-Vendor 3D E-Commerce

[Live](#) · [GitHub](#)

[React](#), [Three.js](#), [Node.js](#), [Socket.io](#), [BullMQ](#), [Redis](#), [Razorpay](#)

- Built a multi-vendor marketplace with 3 role-based dashboards — automated order splits, Razorpay webhook verification, PDF invoices, and delayed escrow payouts via BullMQ.
- Real-time delivery tracking with Socket.io + Leaflet maps, heavy workloads offloaded to BullMQ and Cloudinary.
- Interactive 3D catalog (Three.js, GSAP ScrollTrigger, Framer Motion) holding **120+ FPS** during complex scroll animations on high-refresh displays.
- Optimized throughput to **635 req/s (50.9ms median, P99: 133ms)** with **100% success** across **1,000 VUs** (Autocannon, Artillery, K6); Redis rate-limiting deflected **3,000+ brute-force attempts**; Clinic.js confirmed a **flat ~50MB heap**; **Lighthouse Best Practices 100/100**.

## ACHIEVEMENTS & LEADERSHIP

- LeetCode: 500+ problems** (300+ Medium), **1,600+ Contest Rating**, **238-day streak**.
- Robu Creato 3.0: 3rd Prize** at a national-level robotics competition with **100+ teams**.
- Technical Executive, Evolve Club NIT Bhopal** — led project teams, workshops, and mentoring.